

Designed by Remus Teodorescu 2026

LECTURE 9.2

Gas Flow

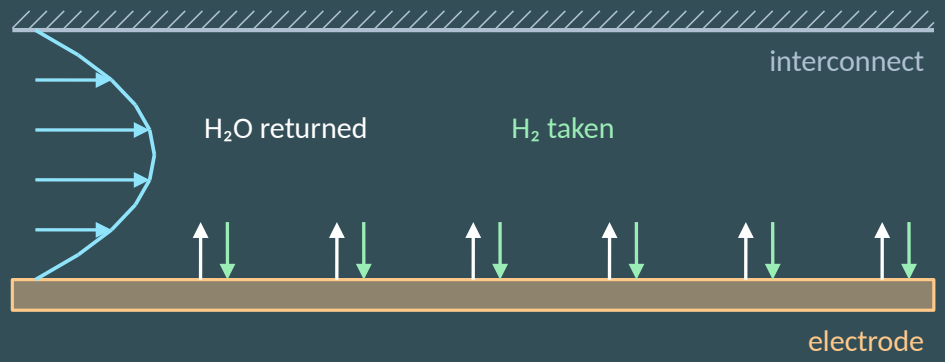
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Deep Learning for Engineering · 2026

GAS · SPECIES · EVERY SPEED



The fuel channel of a solid oxide cell. Three facts make its gas behave like the liquid of the last lecture, so the flow is known; the question is what the gas carries - hydrogen in, steam out - and one PINN, with the speed as an input, answers it for every speed and finds the speed a design asks for.

Learning Objectives

SECTIONS, IN ORDER

1. **The gas** — slow, one for one, Poiseuille from the inlet

2. **What the gas carries** — the species equation, its solution

3. **Every speed** — one PINN, and the speed a design needs

White carries the gas, Incropera the mass transfer and Liu the method; the example is Ex_09.2's fuel channel; L10.2 adds the electrode behind it.

READ, AND WHAT EACH ANSWERS

White ch. 9	when is a gas incompressible?
Incropera ch. 14	how is a species carried?
Liu ch. 2.2	what does an input speed buy?

The Fuel Channel

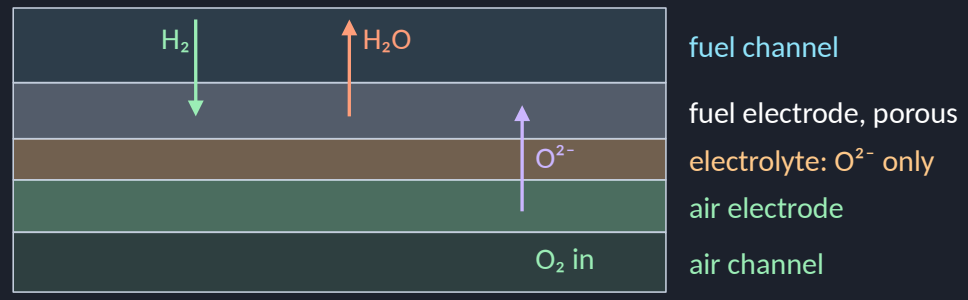
Fuel channel — the gap between a solid oxide cell's fuel electrode and its interconnect. Oxide ions crossing from the air side burn the hydrogen in the porous electrode, and the steam made returns to the gas: one molecule each way for every two electrons.

THE REACTION, AND THE FLUX IT SETS

$$N = \frac{i}{2F} \qquad \mathrm{H_2 + O^{2-} \rightarrow H_2O + 2e^-}$$

N molecules exchanged per area and second, i the current density, F Faraday's constant

BEHIND THE WALL: THE FUEL CELL



THE CHANNEL OF EX_09.2

	value
channel	100 mm long, 1 mm high
gas in	97 % H_2, 3 % H_2O
state	800 °C, 1 atm: 11.36 mol/m³
current	0.5 A/cm²: N = 0.0259 mol/(m² s)
inlet speed	0.3 to 1.5 m/s
run backwards	SOEC: 90 % steam fed, flux reversed

Seen from the gas only; L10.2 adds the electrode. Driven by a supply the same cell splits steam and returns hydrogen: for the channel only the flux's sign changes (MP9.2A). The diffusivity, 8.0e-04 m²/s, is an estimate.

Gas That Flows Like a Liquid

Constant molar density — hydrogen is compressible, but nothing here compresses it: the pressure drop driving the flow is 10^{-7} of the absolute pressure. The wall returns one H_2O per H_2 taken, so the moles per volume stay p/RT, and the velocity; only the mass density changes.

THE MACH NUMBER, AND AN IDEAL GAS

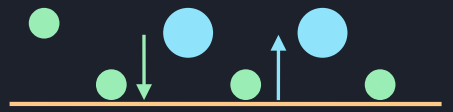
$$Ma = \frac{u}{a}, \quad a = \sqrt{\frac{\gamma RT}{M}}, \quad \frac{\Delta\rho}{\rho} \approx \frac{Ma^2}{2} \qquad c = \frac{p}{RT}$$

a the speed of sound, *γ* the ratio of heat capacities, *M* the molar mass
c moles per volume, *p* pressure, *R* the gas constant, *T* temperature

TALK · 2 MIN How would you explain, in your own words, why the reaction does not push the gas faster? Then, in the drawing on the right, what would change if two molecules entered for each one that left?

AT 800 °C, AND ONE H_2O FOR EACH H_2

quantity	value
speed of sound a	about 2240 m/s
Mach number	below 10^{-3}
density change, from p	about 10^{-7}
moles per volume c	11.36 mol/m³, kept by the wall



H_2 taken, 2 g/mol

H_2O back, 18 g/mol

moles kept, mass not

Light H_2 leaves, heavy H_2O arrives: the mass density rises 2.5 times along the channel at 1 m/s, the moles and the velocity do not. So the species is counted in mole fractions.

Poiseuille From the Inlet

Developed channel flow — the flow between two plates far from the inlet: the parabola of L9.1 in a flat channel. Here Re is 1.3; so low, the entrance stops shrinking with Re and settles at about 0.6 hydraulic diameters, 1.3 mm, 1 % of the channel: the velocity is known before anything is solved.

THE PARABOLA, AND ITS PRESSURE DROP

$$u(z) = 6\,\bar{u}\,\frac{z}{h}\left(1 - \frac{z}{h}\right), \qquad \Delta p = \frac{12\,\mu\,\bar{u}\,L}{h^2}$$

z across the channel, h its height, L its length, $\bar{u}$ the mean speed, μ viscosity

ACROSS THE CHANNEL



zero on both walls

1.5 times the mean in the middle

formed within 0.06 mm

THREE FACTS, THREE NUMBERS

fact	because
constant molar density	Mach below 10^{-3}
velocity unchanged	one molecule for one
parabola from the inlet	Re 1.3, entrance about 1 mm

Poiseuille's pressure drop over the channel is 26 Pa at 1 m/s. Darcy's law of L9.1 with $K = h^2/12$: the flow is solved, and the rest of the lecture is what it carries.

Convection–Diffusion

Convection–diffusion — the equation for a species carried by a known flow: the flow moves it along, diffusion spreads it from rich to poor. Its conditions are the inlet's composition, the flux the electrode returns, and an impermeable interconnect.

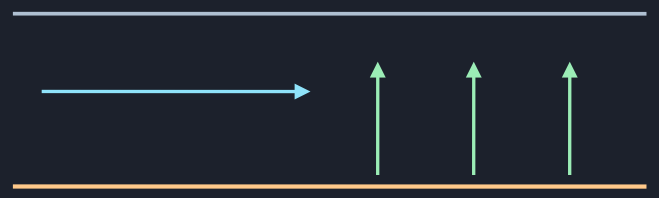
THE STEAM FRACTION, AND ITS THREE CONDITIONS

$$u(z)\,y_x = D\,y_{zz}$$

$$y = y_{in} \text{ at } x = 0, \quad -Dc\,y_z = N \text{ at } z = 0, \quad y_z = 0 \text{ at } z = h$$

y the steam fraction, *x* along, *z* across, *D* the diffusivity, *N* the wall flux

THE TWO WAYS A SPECIES MOVES



along: carried by the flow

across: spread by diffusion

fed at the electrode

ALONG AGAINST ACROSS

$$Pe_L = \frac{\bar{u}L}{D} = 125$$

along	carried, Péclet 125
across	spread, Péclet 1.25
dropped	diffusion along: coefficient $(h/L)^2$

The Péclet number compares carrying with spreading. Along the channel the flow wins by a hundred and twenty-five; across its height diffusion keeps up.

Marching Along the Channel

Parabolic equation — an equation with a direction in which information travels one way only. Without diffusion along it, the species equation is parabolic in x: it marches from the inlet as the heat equation of L8.2 marches in time, and it needs no condition at the outlet.

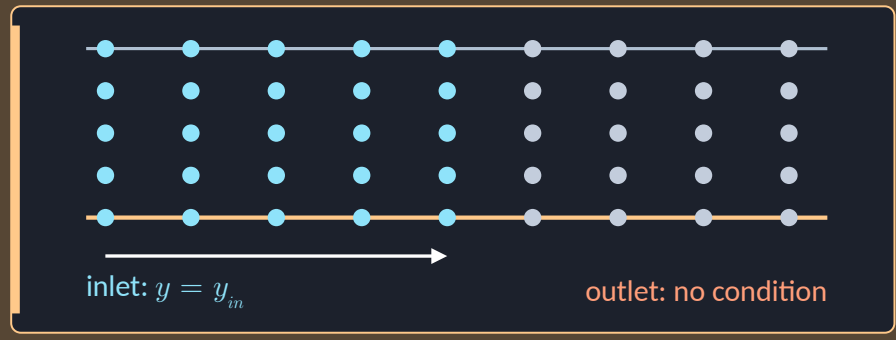
THE SCALED EQUATION, MARCHED

$$6Z(1-Z)\,G\,s\,\theta_X = \theta_{ZZ}, \qquad G = \frac{u_{ref}h^2}{DL}$$

$X = x/L$, $Z = z/h$, $s = \bar{u}/u_{ref}$, θ the rise of y in units of the channel's rise at u_{ref}

TALK · 2 MIN How would you decide, in your own words, which conditions an equation needs and where? Then, in the drawing on the right, why would a zero-slope condition at the outlet be wrong for this channel?

ONE COLUMN AT A TIME



FDM: Crank–Nicolson along, two implicit steps first, central differences across; 401 × 41 nodes meet 0.1 percentage point at every speed in a few ms. The mean keeps rising to the last millimetre, so a zero-slope outlet would be wrong.

Fully Developed Concentration

Fully developed concentration — the state beyond a short entrance in which the profile across the channel keeps its shape while the whole of it rises along. The mean rises by a balance: what the wall returns, the flow carries away. At the inlet the profile is still flat: not the solution there.

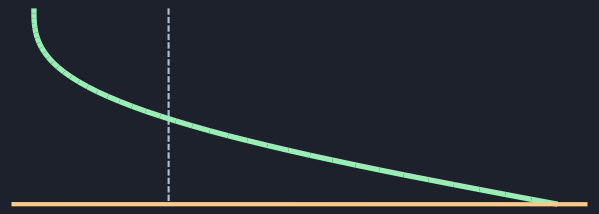
THE BALANCE, AND THE FIELD BEYOND THE ENTRANCE

$$\bar{y}(x) = y_{in} + \frac{Nx}{\bar{u}hc}$$

$$\theta = \frac{X}{s} + G\,\varphi(Z), \quad \varphi = Z^3 - \frac{Z^4}{2} - Z + \frac{13}{35}$$

$\bar{y}$ the mixing cup mean, X/s that balance, φ the profile across, zero mean

ACROSS THE CHANNEL, AT ANY x

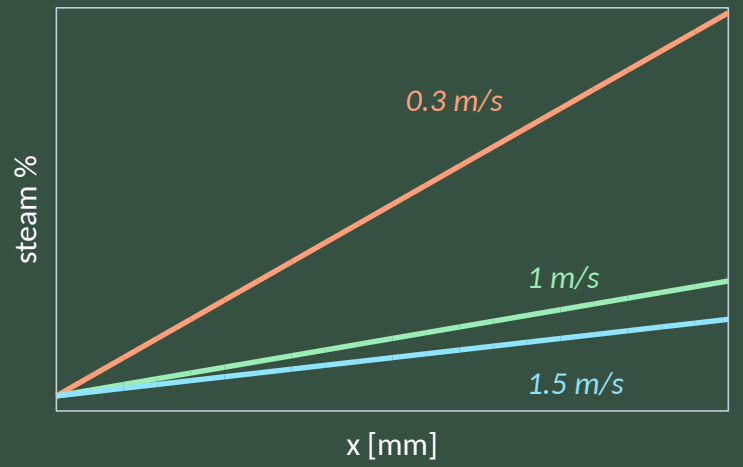


interconnect: a little less

dashed: the mean

electrode: $13G/35$ above it

STEAM ALONG THE CHANNEL



Ex_09.2's field beyond the entrance, scored from 5 mm on: 3 % to 26 % at 1 m/s, to 79 % at 0.3 m/s. The electrode sees 0.11 percentage points more than the mean.

The Sherwood Number

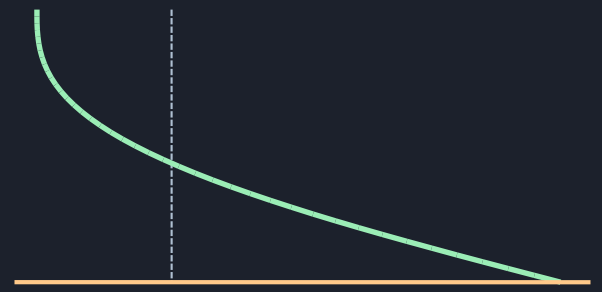
Sherwood number — the mass-transfer coefficient between a wall and the mean of the gas, made dimensionless with the channel's size and the diffusivity. It is the Nusselt number of heat transfer for a species: in a fully developed channel it is a constant of the geometry.

THE WALL AGAINST THE MEAN

$$Sh = \frac{h_m\, 2h}{D} = \frac{2 \cdot 35}{13} = 5.385$$

h_m the mass transfer coefficient, $2h$ the hydraulic diameter, D the diffusivity

THE GAP IT MEASURES



interconnect: a little less

dashed: the mean

electrode: $13G/35$ above it

WHAT EACH METHOD FINDS

	Sherwood
exact	5.385
FDM, 401 × 41	5.38
PINN	5.38 to 5.39
heat, same plates	Nusselt 5.385

One wall at a constant flux, the other impermeable: Shah & London's constant, and Incropera's table 8.1 for heat. Both methods find it to 0.3 % - once the wall value and the mean are read from the same computed column. The excess is a tenth of a per cent of the field.

One PINN for Every Flow Rate

Parametric PINN — a PINN that takes a parameter of the problem as an input beside the coordinates, here the inlet speed s. It is trained once over a range of speeds, and after that any speed is read off, not solved again: the one thing a march cannot do.

HAND CALCULATION BUILT IN, BALANCE HELD BY A LOSS

$$\hat{\theta}(X, Z, s) = \frac{X}{s} + G\,\mathcal{N}(X, Z, s)$$

X/s the hand calculation, built in, $\mathcal{N}$ the network: the profile in units of G, its flow-weighted mean held at 0

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=3, n_hidden=32, n_layers=4)       # X, Z and the speed s in
th = X / s + G * net(torch.cat([X, Z, s], 1))    # the hand calculation, built in
thX = grad(th, X); thZ = grad(th, Z)             # derivatives by autograd
r = (grad(thZ, Z) - 6*Z*(1 - Z)*G*s*thX) / G     # residual, of order one
bal = (w_q * net(...)).sum(1).pow(2).mean()      # balance: mean of N held at 0
```

FDM AND PINN, SCORED EXACTLY

	FDM	PINN
worst error	1e-04, 401 × 41	2 to 6e-06
at the PINN's error	no grid tested	-
the field	one march	one evaluation
new speed	a new march	read off
offline	none	minutes, once

With the mean held by a loss term the PINN is 25 to 120 times more exact than FDM on 401 × 41. Online, same device, no grid tested matches it; against the finest, the field is a few times faster and one number hundreds. Training offline, apart. Numbers: Ex_09.2 section 5.

Design by Gradient

Design by gradient — finding the parameter a design rule asks for from a PINN that has it as an input: the PINN gives the quantity, automatic differentiation its derivative with respect to the parameter, and Newton's method the value, with no further solve.

NEWTON THROUGH THE PINN

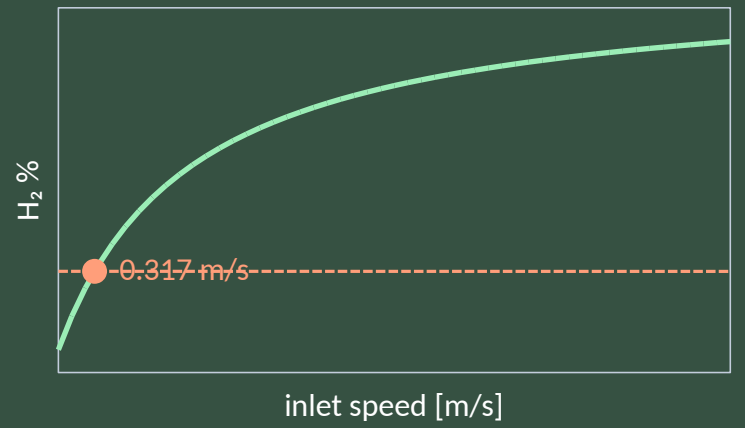
$$s_{k+1} = s_k - \frac{f(s_k) - f^*}{f'(s_k)}$$

$f(s)$ the hydrogen at the electrode at the outlet, $f^* = 25\%$, f' by autograd

THE DESIGN SPEED IN PYTORCH

```
s = (1 / theta_hand).requires_grad_()          # start: the hand calculation
for k in range(5):
    f = h2_at_electrode_outlet(net, s)         # the network, at X = 1, Z = 0
    s = s - (f - 0.25) / grad(f, s)            # Newton, the slope by autograd
```

25 % HYDROGEN AT THE OUTLET



PINN	0.3173 m/s, five steps
exact	0.3174 m/s
then	74 % used, 8 Pa

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
turbulence	eddies of every size	White ch. 6	every fast flow
turbulence models	the mean, and a model for the rest	White ch. 6	industrial CFD
Navier–Stokes problem	can a smooth flow break down?	White ch. 4	the mathematics
the Graetz entrance	the first millimetre of the profile	Incropera ch. 8	short channels
several species	each diffusing through the others	Incropera ch. 14	the air side, MP9.2B
compressible gas	density with the pressure	White ch. 9	Mach above 0.3

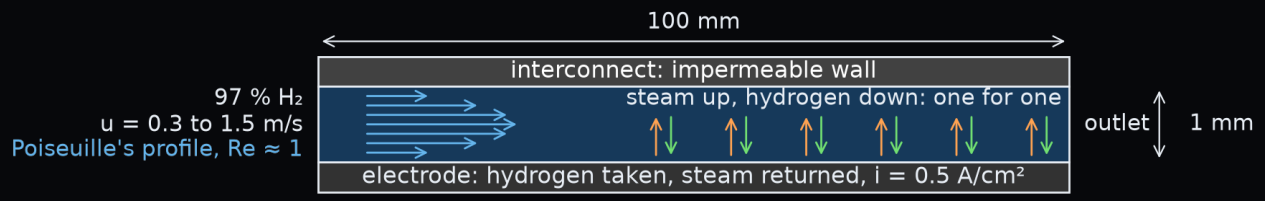
WHY THEY ARE LEFT OUT

The fuel channel's gas is slow, laminar and fully developed, with an exact solution to score against, so none of them is needed to do Ex_09.2. The first three rows are turbulence, which no flow in this cell reaches.

Exercise 9.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the fuel channel, its data, the exact field
- **4** — finite differences, marched from the inlet
- **5** — one PINN for every speed; timed at equal accuracy
- **6** — the inlet speed a design asks for
- **7 – 9** — the report; mini projects MP9.2A and MP9.2B



THE QUESTION EACH PART ANSWERS

1 – 3	why is the gas simple?
4	how fine must the march be?
5	what does a PINN add?
6	which speed keeps 25 %?

REPORT QUESTION, TO CONCLUDE

With its balance held by a loss term the PINN was more exact than any grid tested and found the design speed without a sweep. For which questions about this cell would you train a PINN, and for which march?

SUMMARY

Key Takeaways

GAS AND SPECIES

what it is	a slow gas carrying a species
function	the steam fraction everywhere
train, tune	exact: balance plus a quartic
vs a liquid	the same flow; moles, not mass
used for	fuel use, the electrode's gas

ONE PINN, EVERY SPEED

what it is	a PINN with the speed as input
function	the field at any speed, read off
train, tune	the balance built in; once
vs a march	faster online, no re-solve
used for	design by gradient

Next — L10.1: the species equation again, inside a battery's particle.

Questions

1 – 2 · A NETWORK FOR THE CHANNEL

1 **How can a network learn the fraction of steam in the fuel channel, along it and across it, at any inlet speed?**

The network takes the position along the channel, the position across it and the inlet speed, and is trained until the species equation holds at points spread over the channel and the range of speeds. The rise of the mean along the channel is known from the hand calculation, so it is built into the form of the solution, and the network learns only the small profile across the channel, scaled by the wall flux so that it is of order one. *Liu ch. 2.2; Ex_09.2 section 5*

2 **Why does the network not keep the mean steam fraction of each cross-section right by itself, and how is it enforced?**

The hand calculation built into the solution gives the right mean, but the network can add a mean of its own, and a species equation satisfied only approximately at the training points does not prevent it. A loss term therefore holds the flow-weighted mean of the network's part across the channel at zero. With it the error falls from about a ten-thousandth to a few millionths of the steam fraction, more exact than any grid tested. *Incropera ch. 8; Ex_09.2 section 5*

3 – 4 · EVERY SPEED AND THE DESIGN

3 **What does a network gain by taking the inlet speed as an input, and how does it compare with finite differences?**

Trained once over a range of speeds, a few minutes offline, it gives the field at a new speed by one evaluation, where finite differences march the channel again. Compared online on the same device, no grid tested reaches its error, and against the finest grid it is a few times faster for the whole field and hundreds of times for one number. Its larger gain is not time but the speed as an input that can be differentiated. *Liu ch. 2.2; Ex_09.2 section 5*

4 **How can the inlet speed that leaves a required hydrogen fraction at the outlet be found with a network?**

Automatic differentiation through the trained network gives the derivative of the hydrogen fraction at the electrode at the outlet with respect to the inlet speed, and Newton's method uses it to find the speed for 25 % hydrogen: 0.3173 m/s against the exact 0.3174, in five steps with no further solve. It must start near the answer: from the hand calculation it converges, from one metre per second it diverges. A march answers the same question only by a sweep. *Liu ch. 2.2; Ex_09.2 section 6*

NEXT

L10.1 — Ionic Diffusion

The species equation again, inside the particle of a battery's electrode, where the measurement is a voltage at its surface.

BEFORE THE NEXT SESSION

read	Incropera ch. 14
run	Ex_09.2, sections 1 to 7
bring	your design speed, and its error
think about	a parameter you would make an input

Block nine is finished: a liquid in a tube, and a gas that carries something along a channel.